Image and Voice Pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Image and Voice Pipelines before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Text as Pieces

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Image and Voice Pipelines trace and record the first step you can explain without help.

Use W2: Trace the worked case for Speech **pipeline**s and multimodal **evidence**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Image and Voice Pipelines explanation against the supplied facts before discussing it.

Use the worked case for Speech **pipeline**s and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Image and Voice Pipelines target to a trace and a justified conclusion.

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Image and Voice Pipelines case and show the deciding evidence.

A voice book helper transcribes speech, extracts preference, then ranks books.

Where can a wrong suggestion begin?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Image and Voice Pipelines adds to Transparent Book Recommender and keep the evidence.

If voice or image **input** is added, show each transformation and a safe fallback.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Image and Voice Pipelines facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Image and Voice Pipelines facts and keep your reasoning.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Image and Voice Pipelines facts and keep your reasoning.

A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Image and Voice Pipelines facts and keep your reasoning.

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Image and Voice Pipelines facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Image and Voice Pipelines

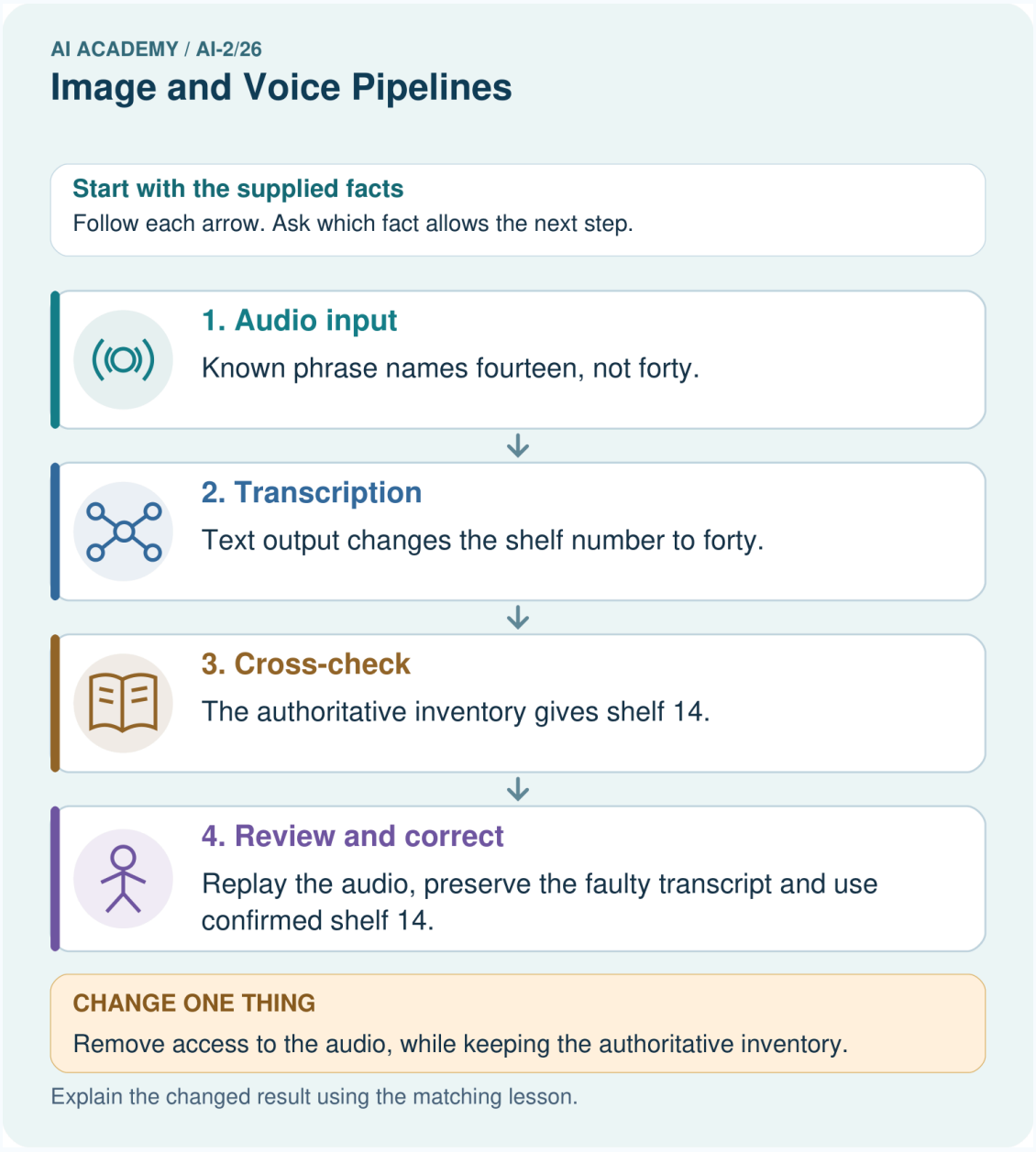
> A fictional librarian dictates a shelf note that passes through transcription and catalogue

> lookup. Explain the labelled step before checking the changed condition.

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which stage first fails, and why should the faulty transcript not be silently overwritten?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

1. Audio **input**

2. Review and correct

3. **Transcription**

4. Cross-check

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Review and correct Replay the audio, preserve the faulty transcript and use confirmed shelf 14. A correct later lookup cannot repair an earlier wrong **input** by magic.

Locate the first difference between known audio and transcript, then compare the stated source authority. Keeping both records makes the correction auditable.

A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove access to the audio, while keeping the authoritative inventory. Which conclusion is supported by the supplied facts?

1. Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

The inventory resolves the shelf task; it does not recreate unavailable audio.

Repair a missing building block

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

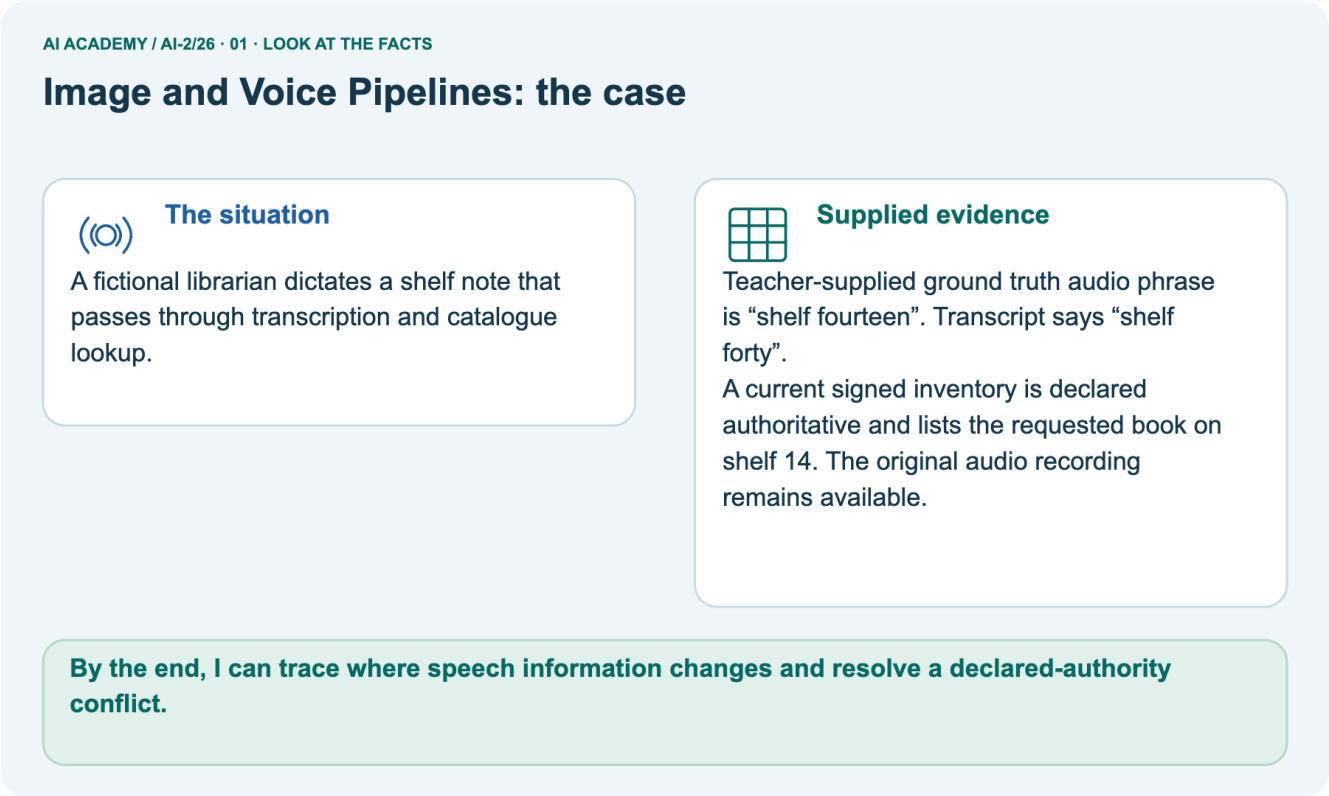
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Image and Voice Pipelines picture from its labels, then record a relevant limit.

01. Image and Voice Pipelines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

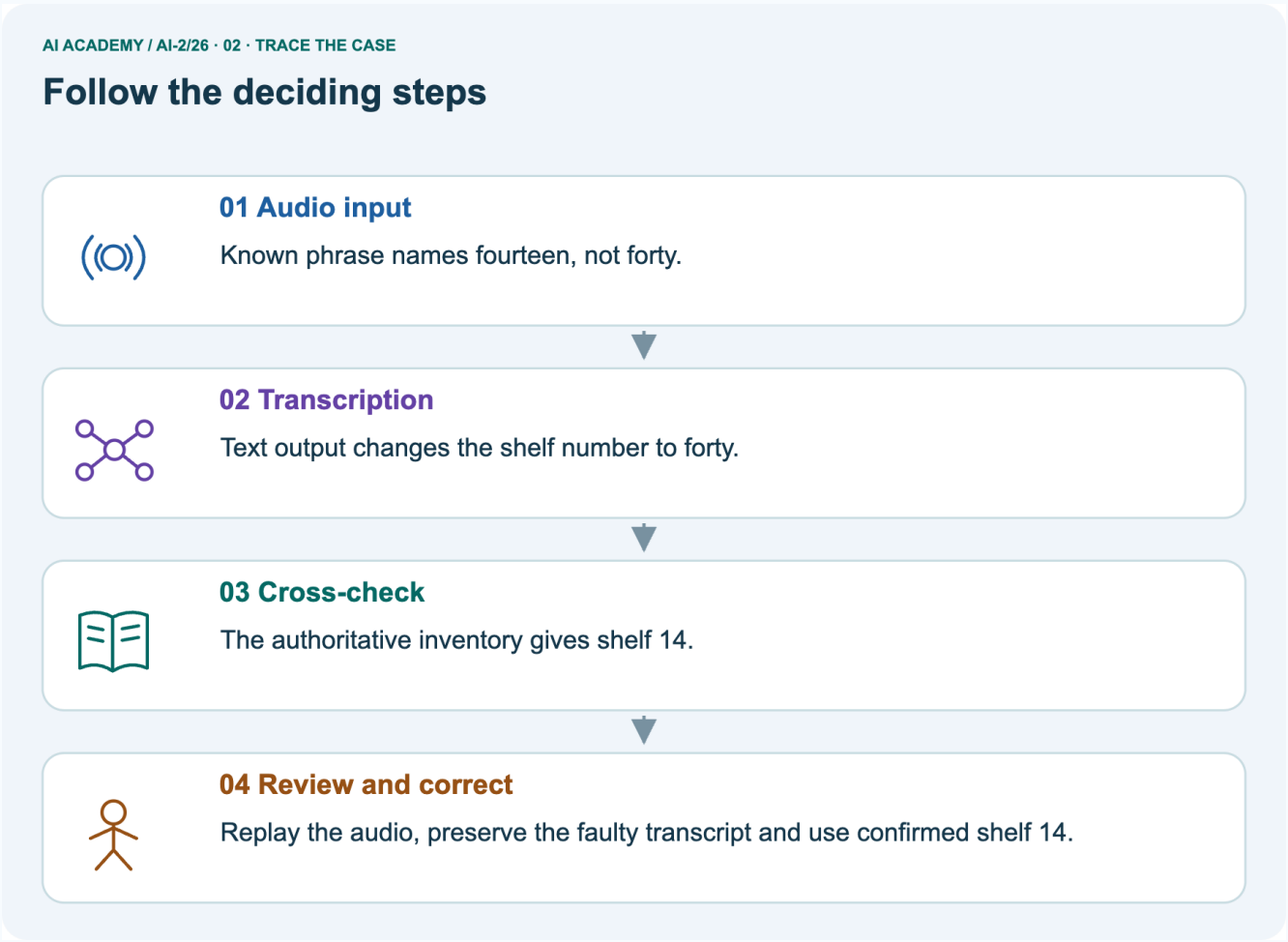
A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup. Supplied facts: Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



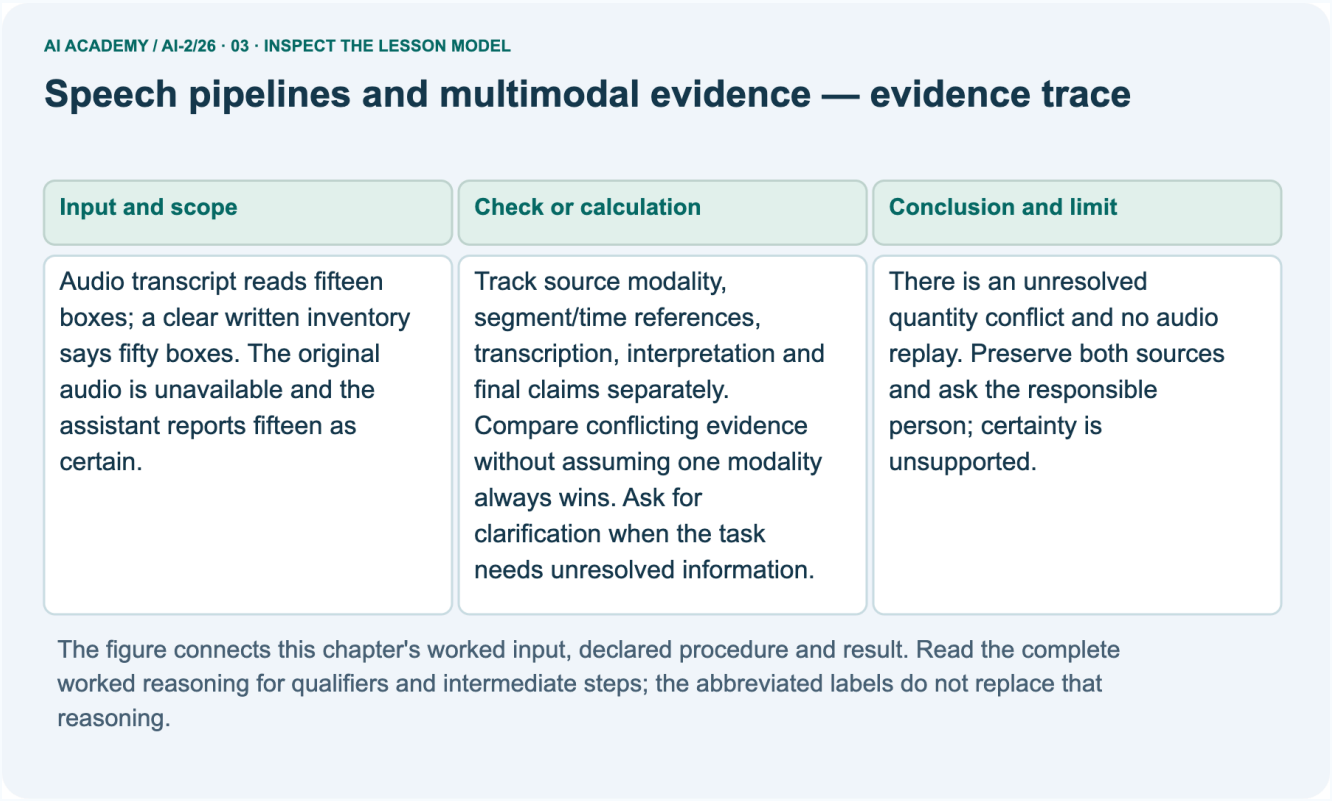
Follow the labelled picture with the matching explanation. Open the vector diagram

Audio **input**: Known phrase names fourteen, not forty. → **Transcription**: Text **output** changes the shelf number to forty. → Cross-check: The authoritative inventory gives shelf 14. → Review and correct: Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech **pipeline**s and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

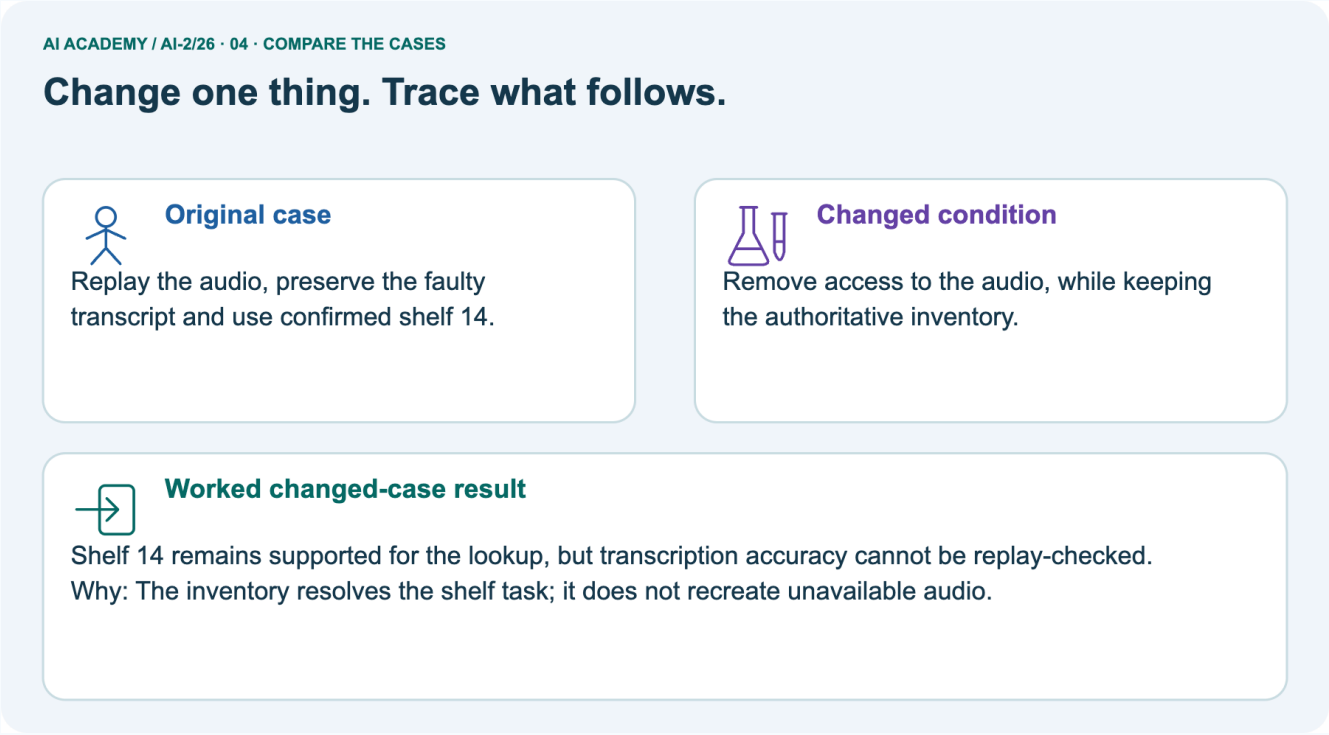
Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

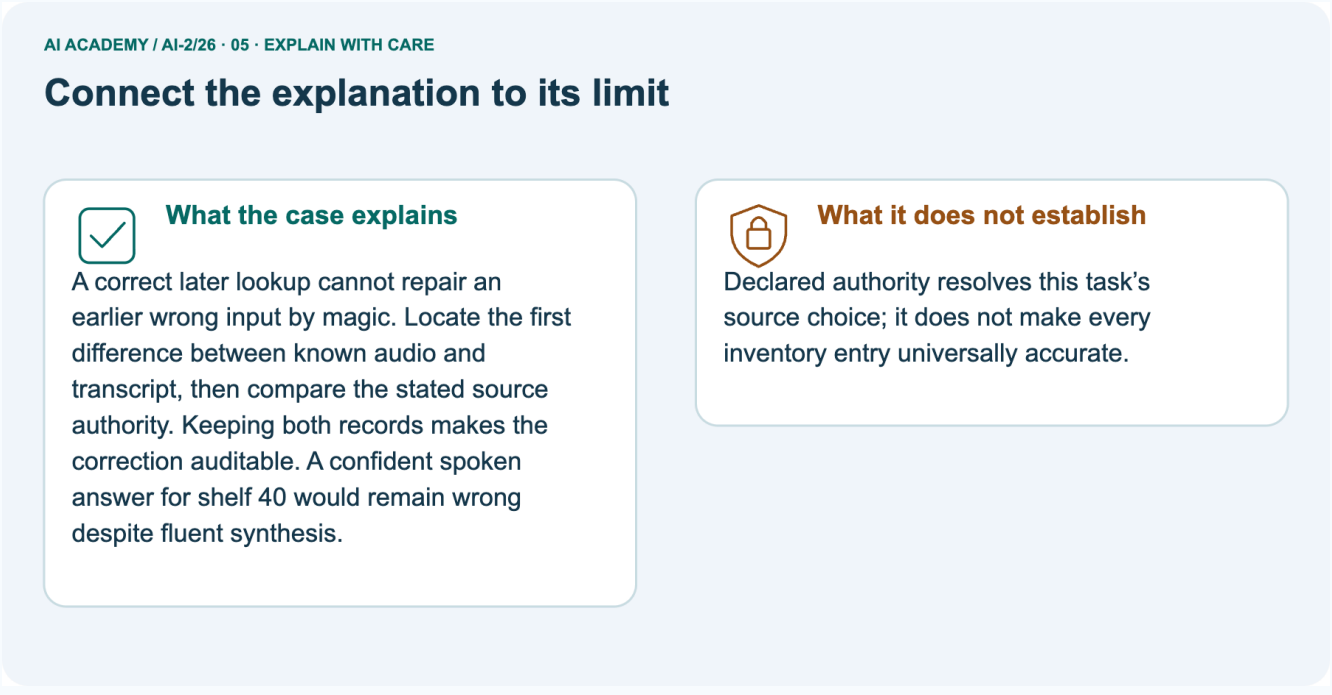
Original case: Replay the audio, preserve the faulty transcript and use confirmed shelf 14. Changed condition: Remove access to the audio, while keeping the authoritative inventory.

Expected result: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked. Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

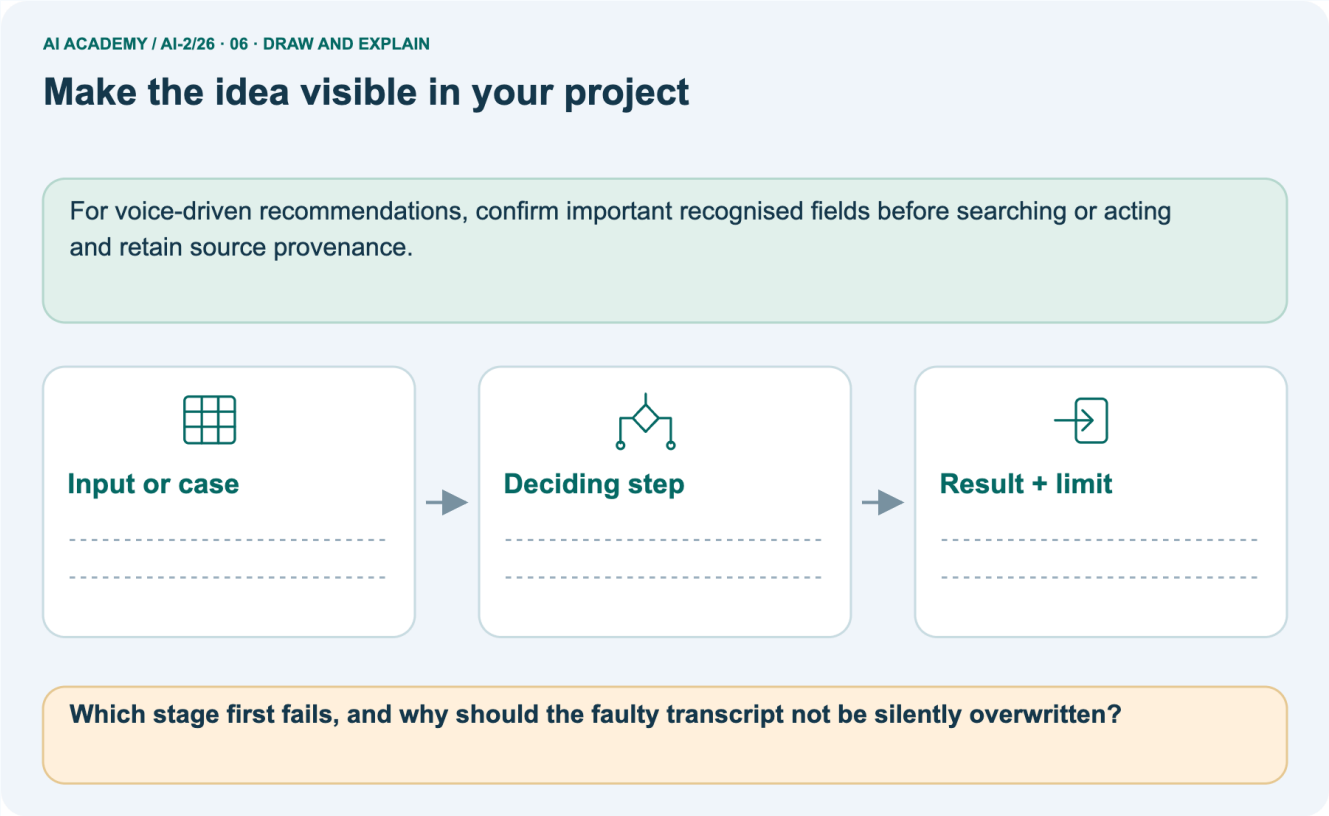
Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Limit: Declared authority resolves this task’s source choice; it does not make every inventory entry universally accurate.

> Explain the case, then point to a limit on the claim.

> **My picture explanation, deciding evidence and one limit**

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which stage first fails, and why should the faulty transcript not be silently overwritten?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.